

Intelligent Voice Bot: End-to-End Architecture & Latency Budget

1. System Overview

An ultra-low-latency real-time voice agent designed for enterprise customer service, orchestrating Automated Speech Recognition (ASR), Large Language Model (LLM) reasoning, and Text-to-Speech (TTS) synthesis.

2. End-to-End Pipeline & Latency Budget

Pipeline Stage	Technology	Latency Target	Optimization Technique
Audio Streaming & VAD	WebRTC + Silero VAD	< 150ms	Real-time chunk streaming with voice activity detection
Speech-to-Text (ASR)	OpenAI Whisper (Base) / Deepgram	< 450ms	Streaming audio transcription with vocabulary boosting
Intent & Dialogue Manager	GPT-3.5-Turbo / Claude Instant	< 600ms	Token streaming, condensed prompt templates, few-shot routing
Text-to-Speech (TTS)	ElevenLabs Turbo v2 / gTTS	< 300ms	Sentence-level chunked audio streaming
Network & Audio Playback	WebSocket Audio Channel	< 100ms	Bi-directional full-duplex audio buffer
Total End-to-End Latency	Integrated Pipeline	< 1600ms	Sub-2-second target achieved

3. Conversational State Machine

- **Greeting State:** Welcomes caller, identifies company context.
- **Intent Triage:** Resolves billing, technical support, or FAQs.
- **Interrupt Handling:** Cancels current audio output if user begins speaking mid-sentence.
- **Safety Fallback:** Immediate human escalation if user distress or high anger is detected.